

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	I	Any two (x1) from A. (Below 50°C) lactase {is more active/ has more kinetic energy} when free (1) B. (Because lactase and lactose can move freely) there is more chance of {successful collisions/ enzyme-substrate complexes forming} (1) C. Takes longer for lactose to diffuse through the alginate beads/ ORA (1)		1	1	2		
			II	A. lactase is more active when immobilised/ comparison of trends of immobilised and free (1) ORA B. the free lactase is {denaturing/denatured} (1) C. Because the immobilisation {stabilises lactase/ maintains shape of active site} (1)		2	1	3		
		(ii)		All give a (blue to orange) colour change with Benedict's so there would be no difference in reactants or products in the reaction/ Reactants and products would both give a blue to orange colour change/ All {monosaccharides/ glucose and galactose} are reducing sugars (1)			1	1		1
				Question 5 total	7	3	4	14	0	1